

STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA
DIPLOMA EXAMINATION (C-24)
C-24-SEP-2025
SEMESTER III, MID-I EXAM



PCODE
243001

AA/CPS

SC-301

Applied Engineering Mathematics

Exam Date: 02-09-2025

Duration: 1 Hour [09:30 AM To 10:30 AM]

Session: FN
[Total Marks: 20]

PART-A

Instructions:

1. Answer the following questions.
2. Each question carries **ONE** mark.

4 X 1 = 4

1. Evaluate: $\int (x^7 + 7^x + 7) dx$.

2. Evaluate: $\int \tan(2x + 1) dx$

3. Evaluate: $\int \frac{dx}{\sqrt{x^2 - 81}}$

4. Evaluate: $\int \frac{dx}{5 - x^2}$.

PART-B

Instructions:

1. Answer the following questions.
2. Each question carries **THREE** marks.

2 X 3 = 6

5(a). Evaluate: $\int \frac{\sin^{-1}x}{\sqrt{1-x^2}} dx$

--- OR ---

5(b). Evaluate: $\int \sin^2 2x dx$.

6(a). Evaluate: $\int \frac{dx}{\sqrt{x^2 + 8x + 25}}$.

--- OR ---

6(b). Evaluate: $\int \frac{x-1}{\sqrt{x^2-4}} dx$.

PART-C

Instructions:

1. Answer the following questions.
2. Each question carries **FIVE** marks.

$$2 \times 5 = 10$$

7(a). Evaluate: $\int \frac{dx}{\sqrt{x+2} + \sqrt{x+3}}.$

----- OR -----

7(b). Evaluate: $\int \sin 7x \cos 5x \, dx.$

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8(a). Evaluate: $\int \sqrt{x^2 + 4x - 12} \, dx.$

----- OR -----

8(b). Evaluate: $\int \frac{dx}{3 + 2 \cos x}.$

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